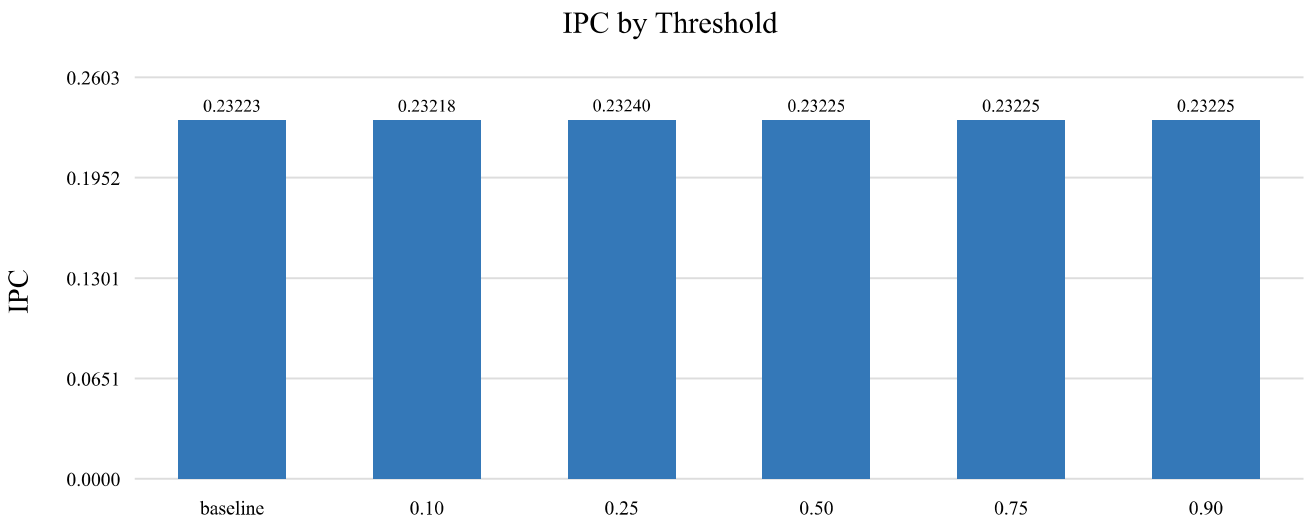
CS683 PA2: Dynamic I-RRIP

Trace1.gz; single-core ChampSim; 25 million warmup and 25 million simulation instructions.

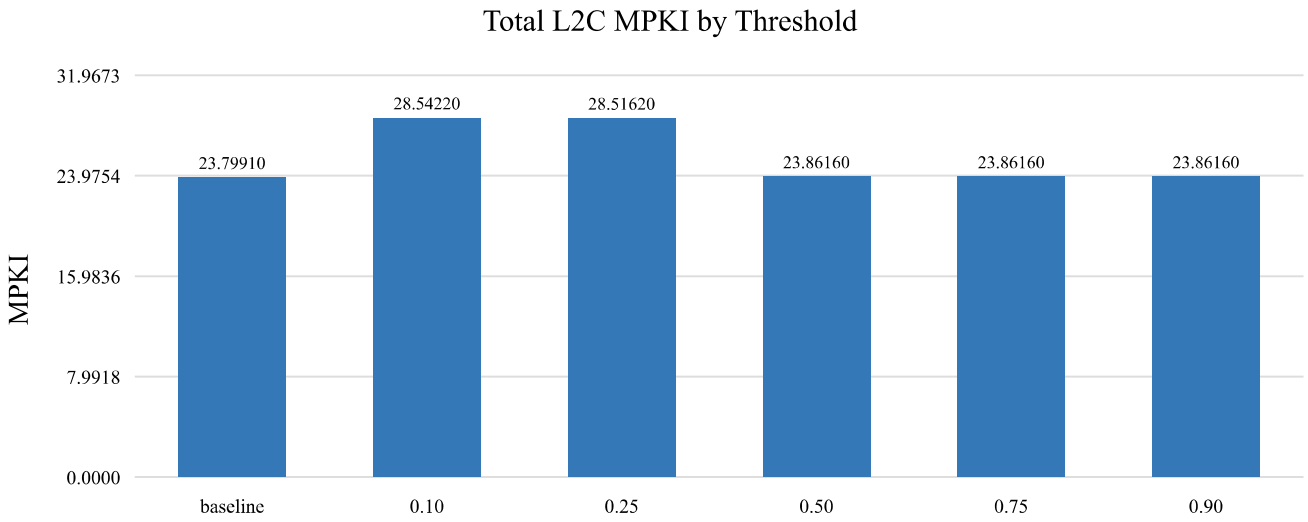
Experimental results

Threshold	IPC	Speedup	Total MPKI	Data-load MPKI	Instruction-load MPKI	Follower policy
baseline	0.232232	1.000000x	23.7991	7.06972	10.5255	SRRIP
0.10	0.232178	0.999767x	28.5422	8.67768	9.66028	I-RRIP
0.25	0.232396	1.000706x	28.5162	8.65824	9.6566	I-RRIP
0.50	0.232250	1.000078x	23.8616	7.09548	10.5098	RRIP
0.75	0.232250	1.000078x	23.8616	7.09548	10.5098	RRIP
0.90	0.232250	1.000078x	23.8616	7.09548	10.5098	RRIP

IPC by threshold



Total L2C MPKI by threshold



Discussion

Baseline SRRIP IPC is 0.232232. Threshold 0.25 has IPC 0.232396, but total L2C MPKI rises from 23.7991 to 28.5162. Instruction-load MPKI falls while data-load MPKI rises, indicating a trade-off rather than a substantial overall performance improvement.

Thresholds 0.50, 0.75 and 0.90 have identical measured results and finish with RRIP as the follower policy. Thresholds 0.10 and 0.25 finish with I-RRIP. Only Trace1 was available for this sweep, so results should not be generalized to other workloads.